Chaebin Jeong

M.S. student in Artificial Intelligence and Data Science, Sejong University, Seoul, Korea
chaibin2000@sju.ac.kr · jng.chaebin@gmail.com · +82 10-8339-6770

RESEARCH INTERESTS

I study where foundation models fail on real-world data and how to repair them: diagnostic evaluation of LLMs and time-series foundation models, knowledge-graph-grounded LLM reasoning and agent memory, and efficient on-device recommendation built on food knowledge graphs.

EDUCATION

M.S. in Artificial Intelligence and Data Science Sep 2025 – Aug 2027 (expected)
Sejong University, Seoul, Korea GPA 4.38/4.50

- Food and Nutrition AI (FNAI) Lab, advised by Prof. Donghyeon Park

B.S. in Electrical Engineering Mar 2020 – Aug 2025
Sejong University, Seoul, Korea

- ABEEK-accredited engineering program (Washington Accord recognized)

PUBLICATIONS

ACCEPTED

[C1] FoodVitals: A Diagnostic Benchmark for LLMs on Nutritional Literacy, Nutrient–Health Relations, and Food Safety
CIKM 2026, Resource Track (to appear) *Second author, equal contribution*
Public benchmark of 1,158 verified items grounded in USDA, EFSA, and FSIS sources, probed in multiple formats (MCQ, Yes/No, short answer, ranking). Zero-shot evaluation of six LLMs shows that single-format probes overstate mastery; e.g., no model ranks four foods by a nutrient above 29% accuracy.

[C2] KARMA: Knowledge Graph-Based Automated Reasoning Materialization and Alignment
Findings of EMNLP 2026 *Second author*
KG-grounded contrastive data synthesis with slot-level preference optimization (SPA) for LLM reasoning.

UNDER REVIEW

[U1] Learning the Rate but Not the Days: Diagnosing and Repairing Time Series Foundation Models on Zero-Inflated E-Commerce Demand
Knowledge-Based Systems (under review) ***First author***
Diagnostic framework that attributes forecast error on zero-inflated demand (37–56% zero days) to four measurable causes. No evaluated zero-shot TSFM (Chronos-2, Chronos-Bolt, Moirai-2, TimesFM-2.5) beats the classical SBA baseline on category-average MSE. Repairing only the fine-tuning objective of a retrieval-augmented TSFM stack closes its 3–20% deficit; the proposed SBA-ZI estimator attains lower pooled CRPS than every evaluated zero-shot TSFM. Released a new 31,318-series item-level panel.

[U2] KEEPER: Learning Control-Sufficient Textual Memory for Long-Horizon LLM Agents
Under review, ACL Rolling Review *Second author*
Defines self-consistency bias, the persistence of incorrect claims in recursively rewritten agent memory, and compares four auxiliary training objectives with offline RL on ScienceWorld and ALFWorld (Llama-3.2-3B). Reward-related prediction yields the highest task performance before and after offline RL.

KOREAN CONFERENCES

[D1] Mitigating Distribution Mismatch in Zero-Inflated Time Series Foundation Models: Mixture of Experts-Based Domain Adaptation and Log Transformation Strategies
KCC 2026 (Korea Computer Congress), poster ***First author***
MoE-module fine-tuning with log transformation, improving MSE by 7.5% on average.

[D2] An Efficient Travel Itinerary Generation Agent Based on LLMs with Plugin Tools and ReAct Orchestration
KICS Winter Conference 2026 *Co-author*

IN PREPARATION

[P1] Substitution-Aware Food Knowledge Graph and MoE-Based On-Device Food Recommendation Agent (working title)
In preparation ***Co-first author***
Food-substitution knowledge graph and a mixture-of-experts recommendation agent for on-device use.

RESEARCH PROJECTS

Project Researcher, KIRD 2026 Global R&D Challenge Jun 2026 – Oct 2026
AdaptEat: Explainable Cross-Cuisine Food Recommendation over a Unified Korea–Uzbekistan Food KG, with New Uzbekistan University

- Team project: a Korea–Uzbekistan food KG from 1,000 recipes (33,001 nodes, 289,036 edges) and a two-route recommender that suggests Korean alternatives to Uzbek dishes.
- Built the flavour feature (FlavorDB profiles for all 1,000 dishes), the KG route (PathSim over shared cooking-step paths), and their reciprocal rank fusion with the feature route: the fused list ranks a same-category dish first for 29.2% of queries (24.0% without the KG route), and its top 3 shares the main ingredient for 21.4% of Uzbek dishes vs. 4.6% at random.
- Built the graph-grounded explanation module, its automatic checker (96.5–97.3% of explanations fully verified against the graph), and its HTML/CSS view in the public web demo; co-authored the team proposal.

Project Researcher, Yanolja Joint Research Sep 2025 – Aug 2026
Industry collaboration: LLM travel-itinerary agent for tourism data

- Built a ReAct travel-planning agent on the TravelPlanner framework: the LLM plans while plug-in tools (attraction, accommodation, restaurant, and distance search; notebook; planner) look up facts, so new data or domains need no retraining. Moved the tools to a Seoul dataset with district-level city indexing.
- Diagnosed why a LoRA/RAFT-tuned Llama 3.1 planner hallucinated budgets and currencies (domain mismatch and overfitting of its Reddit fine-tuning data) and redesigned the roles: a stronger model handles reasoning and tool choice, and the planner may use only retrieved information. Fixed agent loops on simple lookups and planner over-calls.
- Co-authored the KICS 2026 paper on this agent.

HONORS AND LEADERSHIP

Student Lab Leader, FNAI Lab, Sejong University Jun 2026 – present
Coordinate lab research activities and student members

Excellence Award, Multicultural Research Team Best Practice Contest Dec 2025

SKILLS

Programming	Python, PyTorch (advanced); SQL, C++, HTML/CSS (familiar)
ML and LLM	Hugging Face Transformers, vLLM; LLM evaluation and benchmark design; LLM agents (ReAct, tool use); knowledge-graph retrieval (PathSim); time-series foundation models (Chronos, Moirai, TimesFM); mixture-of-experts fine-tuning; probabilistic forecasting (CRPS)
Tools	Git, LaTeX
Languages	Korean (native), English (professional working proficiency)